CURRICULUM VITAE

Muhammad Abu Bakar

Geospatial Data Specialist



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Professional experience

11/2025 – present

Geospatial Data Specialist · Beagle Systems, Hamburg

Beagle Systems flies long-range drones beyond visual line of sight (BVLOS) to monitor pipelines, utilities and other infrastructure.

- Developed an open-source drone LiDAR pipeline that georeferences point clouds with PPK, classifies power-line scenes, rebuilds missing conductors from their sag and rates nearby vegetation by risk; tree-fall analysis in progress.
- Built an AI-driven right-of-way and construction-site monitoring pipeline: deep-feature-matching georeferencing, a fine-tuned foundation model for detection, an open-source VLM to verify it, and real-time inference on an NVIDIA Jetson.
- Fine-tuned a building-segmentation foundation model on in-house labels created from 10 cm orthophotos and scaled it across the Cologne area.
- Built a PostGIS and TiTiler backend that serves large GeoTIFFs to the frontend as map tiles, and supported flight planning with pre-flight checks and new geodatabase data.

03/2024 – 10/2025

Geospatial (EO) Data Scientist, part-time · Karuna Technology UG, Germany

- Fine-tuned a pretrained deep-learning model on in-house training data to map urban tree canopies from aerial imagery, then tracked their health with Planet NDVI and NDWI time series in Google Earth Engine to flag underperforming trees.
- Projected the long-term ecosystem services of the mapped trees through a tree-benefits API, from carbon storage to further environmental benefits, estimating the urban canopy's value over the coming decades.
- Built Google Earth Engine workflows that clean multi-year Sentinel-1 radar time series with speckle filtering and apply an equation to derive soil-moisture maps, showing where golf courses suffer from water stress.

04/2021 – 02/2024

GIS Specialist & Data Engineer · MapTransfer, Germany

- Managed the end-to-end geodata workflow for a construction company: georeferenced construction plans and orthophotos, converted them into cloud-optimized GeoTIFFs, uploaded them to AWS S3, and set up QGIS with PostgreSQL as the customer's secure front end.
- Designed and authored several GIS courses for students and professionals, covering LiDAR processing, spatial analysis, Spatial SQL, ALKIS and GeoServer.
- Converted AutoCAD data in QGIS into accurate, ALKIS-compliant GIS layers for telecom infrastructure projects.
- Built interactive Web GIS applications in OpenLayers, serving WMS layers through GeoServer from PostgreSQL.

Education

2017 – 2020

MSc Geology · University of Freiburg, Germany

- Thesis: Simulation of multispectral and hyperspectral imaging techniques for mapping of epithermal ore deposits in vegetated zones.

2016 – 2020	**MSc Remote Sensing & Geographic Information Systems** · National University of Sciences and Technology, Pakistan ○ Thesis: Development of an open-source Python-based slope stability model that incorporates remote sensing and geotechnical data for slope failure assessment.
2012 – 2015	**BSc Geology** · Bahria University, Pakistan ○ Thesis: Petrophysical analysis of Joya Mair – Well 4.

Academic projects

Landsat · Sentinel-2	Glacier mass balance of Mýrdalsjökull from historical imagery and DEMs: the melting rate doubled after 2000.
MODIS · TRMM	Snow cover and precipitation variability, 2000–2015.
Landsat 7	Snow-cover change in the Kabul Basin and its impact on water resources.
Landsat 8 · ASTER	Land cover, erosion and morphometry of Mauna Kea.

Further training

2022	NASA Applied Remote Sensing Training (ARSET): five trainings on SAR-based disaster assessment, flood monitoring and modeling, crop mapping with polarimetric SAR and optical data, air-quality model output, and pre- and post-fire monitoring. UNAVCO / GAGE short course: InSAR processing and time-series analysis for geophysical applications (ISCE, ARIA-tools, MintPy).

Skills

Machine learning	Fine-tuned foundation models, open-source LLMs and VLMs (self-hosted, vLLM), deep feature matching, CNNs, CatBoost; PyTorch, TensorFlow; edge inference on NVIDIA Jetson
LiDAR	PDAL, CloudCompare, WhiteboxTools; PPK with RTKLIB, point-cloud classification, DEM and DSM generation
Remote sensing	Multispectral, hyperspectral and SAR data; ENVI, ERDAS Imagine, eCognition, ESA SNAP
GIS	QGIS, ArcGIS Pro
Programming	Python (GDAL, Rasterio, Fiona, scikit-learn, Cartopy), R, Spatial SQL, Bash
Databases	PostgreSQL/PostGIS, SpatiaLite, DuckDB
Cloud and DevOps	Google Earth Engine, AWS (EC2, S3), Docker
Web mapping	OpenLayers, Leaflet, GeoServer, TiTiler
Geology	Numerical modeling, petrophysics
Other	Git, LaTeX

Languages

English	C2
German	B2

Freiburg im Breisgau, 14 October 2026

Signature